
NSF FUTURE MANUFACTURING DATA CHALLENGE | FINAL SUBMISSION

Predicting local DED track width from thermal history

Hierarchical condition baselines, local residuals, and nested four-track evaluation

Team Submission

July 31, 2026 | Nested four-track outer validation

Can thermal history predict local geometry after printing?

A DED track is not a uniform line: width and both boundaries vary along x.

Thermal frames record the moving melt pool; profilometry records final geometry.

The modeling task maps a frame and its causal history to the matching local segment.

A credible result must generalize across laser power and quantify uncertainty.

OUR TASK

Predict local width and boundary position from thermal history

Ground truth: center, width, and boundaries from profilometry

Inputs: thermal history; post-process SEM tested separately

Evaluation: four nested leave-one-track-out outer tests

Three modalities, one physical x-axis, four untouched outer tests

THERMAL VIDEO

400 active frames per track

400 active frames per track at 50 fps.

10 mm/s scan speed gives 0.2 mm per frame.

Size, temperature, gradients, and asymmetry.

Causal history lives in the TIME domain.

SEM IMAGERY

Post-process SEM ablation

Available imagery is post-process.

The processed center band is masked.

Flank texture is tested, not assumed causal.

Selected in zero outer folds.

3D PROFILOMETRY

Local width and boundaries

Robust cross-track substrate detrending.

Connected 30%-height boundary crossings.

0.40 mm physical smoothing window.

Missing acquisition gaps remain excluded.

Every track is untouched once; model, feature family, preprocessing, and calibration are selected only inside the other three tracks.

From raw frames to leakage-safe nested predictions

1

Extract

Detrend each cross-section and find connected left/right crossings

**2**

Align

Use physical x in mm; never normalized row position

**3**

Decompose

Separate condition baseline from local center/log-width residuals

**4**

Test

Hold out each track once; select candidates inside the other three

Audit rule: geometry, feature choice, model choice, and uncertainty calibration never see the current outer track. Generative AI assisted coding and layout; it did not alter measurements.

Ground truth first: robust boundaries, then physical alignment

STEP 1 - LOCAL GEOMETRY EXTRACTION

Measure one connected bead in every valid cross-section

Interpolate missing y pixels only within each cross-section.

Fit substrate slope outside the bead shoulders.

Locate connected 30%-of-peak crossings around the central maximum.

Smooth left and right boundaries over 0.40 mm.

STEP 2 - PHYSICAL TIME-SPACE ALIGNMENT

Match each thermal frame to measured x in millimeters

Detect laser shutoff; retain the prior 400 active frames.

Extract hot components above a fixed 1500 K threshold.

Convert time to x at 10 mm/s and 50 fps.

Reject frames more than 0.10 mm from valid profilometry.

Separate condition width from local variation

CAUSAL MULTISCALE HISTORY

Describe the pool now, then summarize how it arrived

Pool axes, temperature, mass, gradients, and asymmetry.

Cooling-tail decay, centroid velocity, and shape change.

5-, 10-, and 20-frame slopes, persistence, and changes.

Robust local normalization isolates within-track departures.

CONSTRAINED NESTED MODEL

Predict a track baseline, then local residuals

Track summaries predict baseline center and log-width.

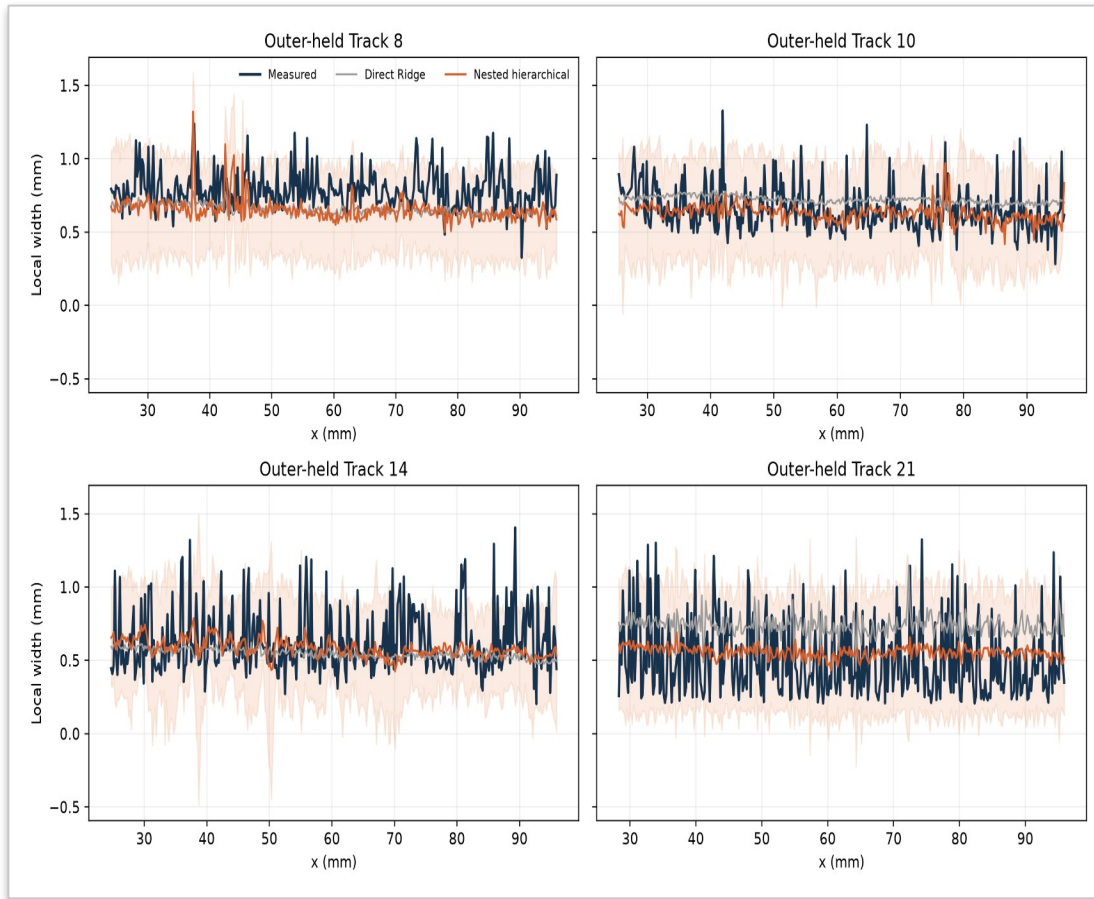
Local history jointly predicts center and log-width residuals.

$\exp(\text{log-width})$ keeps width positive; boundaries stay ordered.

Nested folds choose Ridge, elastic net, PLS, spline, or GP.

Four-track MAE falls 13.1% while boundaries improve

EXHIBIT 1 - FOUR UNTOUCHED OUTER TRACKS



163 μm
Track-balanced MAE

13.1% lower
vs. direct Ridge

Track-balanced width MAE: 0.187 $\rightarrow$ 0.163 mm.

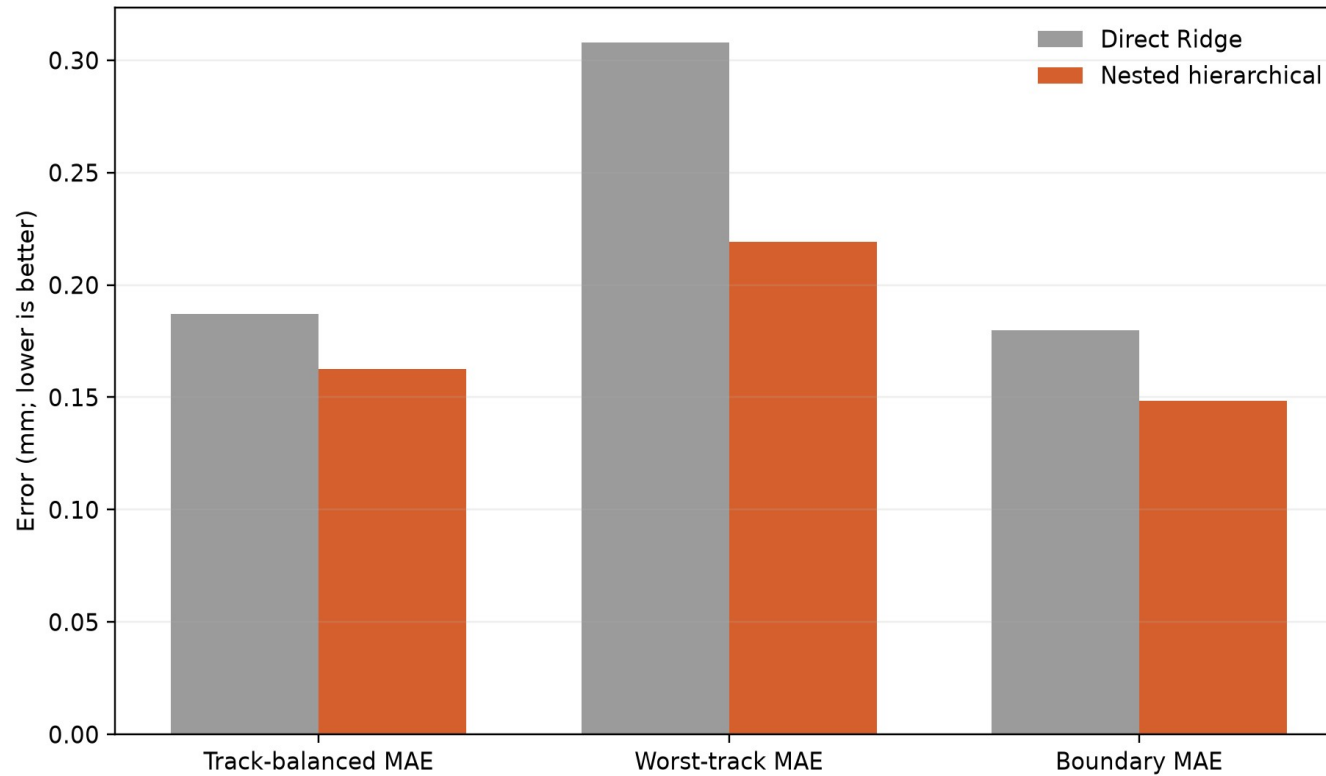
Worst-track MAE: 0.308 $\rightarrow$ 0.219 mm.

Mean boundary MAE: 0.180 $\rightarrow$ 0.148 mm.

Residual correlation: 0.055 $\rightarrow$ 0.124.

Local scaling recovers variation and sharpens uncertainty

EXHIBIT 2 - BEFORE / AFTER SCORECARD



WHAT CHANGED

Condition baseline + local residual wins

1. Predicted variation grows from 21% to 36% of measured scale.
2. Conditional intervals cover 91.4% at 0.738 mm mean width.
3. Global intervals cover 94.2% but are wider at 0.824 mm.
4. Width expands from 0.610 mm easy to 0.912 mm hard.

Improved, but not ready for closed-loop control

Track-balanced R^2 is -0.34; local predictions still miss structure.

Four tracks cannot establish stability across plates or recipes.

Nested Track 21 MAE is 0.219 mm; the prior 0.139 mm is historical only.

Post-process SEM is selected in zero folds.

No causal substrate claim is supported.

WHAT WOULD CLOSE THE GAP

Add more powers, plates, repeats, and substrate pre-scans.

Register SEM before processing instead of inferring substrate from post-scan tiles.

Calibrate intervals by laser-power regime or hierarchical conformal methods.

Predict a full contour distribution, not only center plus width.

Separate the condition, then learn the variation

1

Four-track accuracy

Width MAE falls 13.1%; worst-track error falls 28.9%; boundary MAE falls 17.6%.

2

Physical consistency

Positive width and one shared center reconstruct ordered left/right boundaries.

3

Useful uncertainty

Conditional intervals reach 91.4% coverage and expand in difficult regions.

“Condition shifts and local variation are different problems.”

Questions? | github.com/alphons3t/nsf-chuds | github.com/joeyperez1-debug/nsfsubmit